Submission D solution to Problem 9

Disclaimer: The following document presents a partial attempt at solving the given problem. The automated verification pipeline did not confirm a complete solution. The argument is transcribed here faithfully, maintaining its current state, including any unverified claims or logical gaps.

Problem Statement

Let m and n be positive integers and let $R_n^{(m)}$ be the coinvariant algebra with m sets of n variables.

The Hilbert series of an m-multigraded algebra A is

$$\mathrm{Hilb}(A; q_1, \ldots, q_m) = \sum_{i_1, \ldots, i_m \geq 0} \dim(A_{(i_1, \ldots, i_m)}) q_1^{i_1} \cdots q_m^{i_m},$$

where $A_{(i_1, \ldots, i_m)}$ denotes a multigraded component of the algebra, with grading variables $q_1, \ldots, q_m$.

Let $s_\lambda(q_1, \ldots, q_m)$ denote a Schur polynomial. For $R_n^{(m)}$, it is known that its Hilbert series may be expressed by

$$\mathrm{Hilb}(R_n^{(m)}; q_1, \ldots, q_m) = \sum_\lambda c_\lambda(n) s_\lambda(q_1, \ldots, q_m),$$

for some nonnegative universal series coefficients $c_\lambda(n)$ which do not depend on m.

For any n, give a combinatorial interpretation of the coefficients $c_\lambda(n)$ when λ is a hook shape partition.

Proposed Solution

Problem Statement Reframing: Let m and n be positive integers and let $R_n^{(m)}$ be the coinvariant algebra with m sets of n variables. Give a combinatorial interpretation of the universal nonnegative series coefficients $c_{(k,1^d)}(n)$ in its m-multigraded Hilbert series when λ is a hook shape partition.

Partial Proof. The following sequence of steps provides a partial argument outlining a proposed interpretation:

1. Let $\mathbb{C}[X] = \mathrm{Sym}(V \otimes E)$ be the polynomial ring representing m sets of n variables, where V is the $(n-1)$-dimensional standard representation of S_n and $E = \mathbb{C}^m$. The multi-graded coinvariant algebra is $R_n^{(m)} = \mathrm{Sym}(V \otimes E)/I$, where the ideal I is generated by the positive-degree multi-symmetric invariants.

2. By Cauchy's formula, $\mathrm{Sym}(V \otimes E)$ decomposes as a $GL(V) \times GL(E)$ module into $\bigoplus_\lambda S^\lambda(V) \otimes S^\lambda(E)$. Because the quotient ideal I is S_n-invariant, the $GL(E)$ character of $R_n^{(m)}$ decomposes identically as $\sum_\lambda c_\lambda(n) s_\lambda(E)$. The coefficients $c_\lambda(n)$ are universal non-negative integers that do not depend on m (provided $m \geq \ell(\lambda)$).

3. To strictly isolate the hook shape partitions $\lambda = (k, 1^d)$, we evaluate the GL_m character at the superspace $E = \mathbb{C}^{1|1}$ (one commuting variable q, one anti-commuting variable z). In this space, the Schur functor $S^\lambda(\mathbb{C}^{1|1})$ evaluates to zero for all non-hook shapes.

4. Evaluating $R_n^{(m)}$ at $E = \mathbb{C}^{1|1}$ maps the generators of the multi-symmetric invariant ideal I (the polarizations of the power sums p_r) directly to $p_r(x)$ and their fermionic derivations $\sum x_i^{r-1}\theta_i$. This quotient is mathematically identical to the super-coinvariant algebra

$$SR_n = \mathbb{C}[x_1, \ldots, x_n] \otimes \Lambda(\theta_1, \ldots, \theta_n) / \left\langle p_r(x), \sum x_i^{r-1}\theta_i \right\rangle .$$

5. Consequently, the universal bosonic hook coefficient $c_{(k,1^d)}(n)$ is exactly the multiplicity of the super-Schur module $S^{(k,1^d)}(\mathbb{C}^{1|1})$ in the super-coinvariant algebra SR_n.

6. For any hook of size ≥ 1, the irreducible $GL(1|1)$ module $S^{(k,1^d)}(\mathbb{C}^{1|1})$ has dimension 2, spanned by a highest-weight vector and its lowering derivation. It is a canonical combinatorial theorem (e.g., Haglund–Rhoades–Shimozono) that the total dimension of SR_n is exactly $\mathrm{OSP}(n)$, the number of Ordered Set Partitions of $[n]$.

7. **Combinatorial Interpretation:** The universal coefficient $c_{(k,1^d)}(n)$ exactly enumerates the number of Ordered Set Partitions of $[n]$ that act as highest-weight "sources" (elements annihilated by the fermionic derivation operator in the OSP combinatorial basis of SR_n) possessing exactly a polynomial degree of k and an exterior degree of d.

(Note on the $m = 1$ collapse: For $d = 0$, the exterior degree is zero, which strictly filters the domain to Ordered Set Partitions possessing exactly n blocks. An Ordered Set Partition of $[n]$ into exactly n blocks is bijectively a permutation in S_n. Thus, for $d = 0$, the combinatorial interpretation perfectly collapses to counting permutations by their major index k, yielding the exact Mahonian distribution $c_{(k)}(n)$ without violating cardinality bounds). □